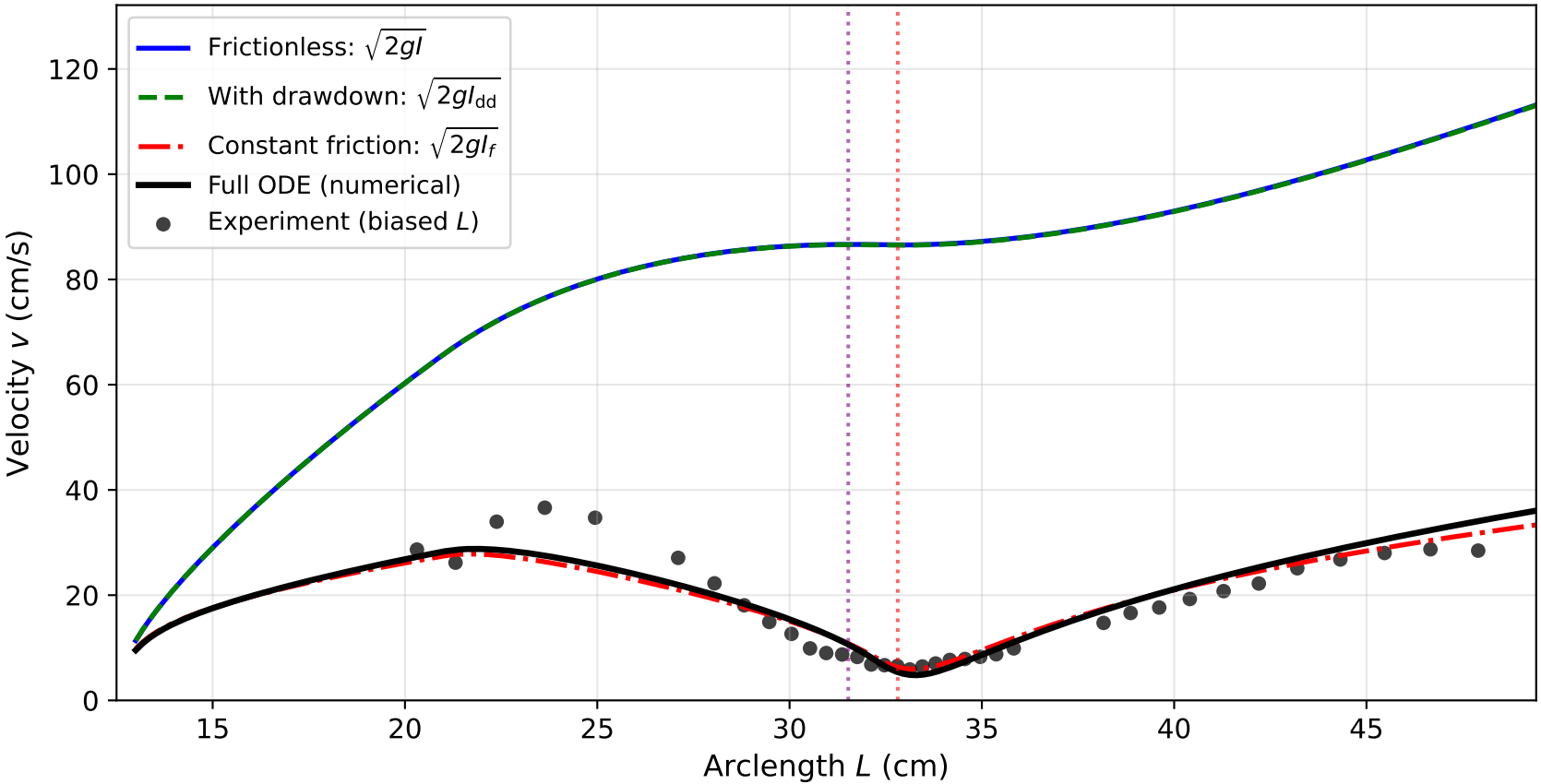
Velocity Comparison: Analytical Bounds vs. Numerical Solution
Frictionless: $\sqrt{2gl}$
With drawdown: $\sqrt{2gl_{\mathrm{dd}}}$
Constant friction: $\sqrt{2gl_f}$
Full ODE (numerical)
Experiment (biased L)
Velocity v (cm/s)
Arclength L (cm)
0
20
40
60
80
100
120
15
20
25
30
35
40
45